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Leveraging big data characteristics for enhanced healthcare fraud detection

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Abstract
This review explores the transformative potential of Big Data analytics in revolutionizing healthcare fraud detection. By examining the 5Vs (Volume, Variety, Velocity, Veracity, and Value), we illustrate how these dimensions can significantly enhance the accuracy and efficiency of fraud detection systems. This review not only analyzes how the 5Vs can be leveraged to improve fraud detection systems but also identifies existing research gaps and proposes future research directions. Furthermore, we address the integration of Artificial Intelligence (AI) and Big Data applications in healthcare fraud detection, highlighting their role in enhancing operational efficiencies and improving the overall quality of healthcare services. This paper aims to provide valuable insights into modernizing healthcare systems through Big Data technology, ultimately contributing to more efficient, reliable, and trustworthy healthcare services.

Keywords Big data · Healthcare fraud · Healthcare systems · 5Vs · Artificial intelligence · Spark · Hadoop

1 Introduction

Healthcare fraud involves fraudulent strategies with the intention of acquiring illegal financial advantages from healthcare programs [1]. These fraudulent activities include invoicing for services that were not provided, misrepresenting the complexity level of procedures, conducting unneeded medical interventions, and falsifying diagnoses. Identifying and preventing healthcare fraud is essential as it results in substantial monetary losses and degrades the standard of patient treatment [2–4]. Traditional techniques for addressing healthcare fraud include manual audits, rule-based systems, and statistical analysis. However, these methods often struggle with the complexity and scale of modern healthcare data [5–7]. Leveraging the Big Data 5Vs–Volume, Variety, Velocity, Veracity, and Value–can help address these issues [8–12]. By analyzing large datasets (Volume), integrating diverse data sources (Variety), processing information in real-time (Velocity), ensuring data accuracy (Veracity), and deriving actionable insights (Value), big data analytics offers promising solutions to detect and prevent fraud effectively.

Deploying big data solutions for tackling fraud in the healthcare industry is a sophisticated and varied task that necessitates an extensive approach [13]. Despite there are significant advantages, organisations must overcome numerous challenges in order to achieve success. One of the key challenges lies in data integration. Healthcare organisations often encounter challenges when integrating various sources of information, such as electronic health records and insurance claims, into a unified system. It is of utmost importance to guarantee the precision of this data, since any errors can result in both false positives and false negatives outcomes, significantly reducing the effectiveness of fraud detection. Scalability is another significant concern. Handling large volumes of data requires robust infrastructure capable of expanding as data grows. Real-time processing poses additional technical demands, as organizations strive to analyze data instantaneously for immediate fraud detection [14]. Privacy and security are paramount considerations, given the sensitive nature of

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patient information [15]. Stringent measures must be in place to protect this data and ensure compliance with healthcare regulations and data protection laws. The cost of implementing these advanced technologies can also be substantial, requiring significant investment in both infrastructure and expertise. Skill shortages further complicate efforts, as finding professionals with the requisite knowledge to develop and maintain big data systems can be a challenge. Despite these challenges, healthcare organisations have the potential to utilise big data technologies in order to improve their efforts in preventing fraud. To fully harness the capabilities of these new technologies and protect the integrity of the healthcare system, organisations have to address the technological, regulatory, and financial challenges.

This review explores the transformative potential of Big Data analytics in modernizing healthcare fraud detection. By examining the 5Vs (Volume, Variety, Velocity, Veracity, and Value), we illustrate how these dimensions can significantly enhance the accuracy and efficiency of fraud detection systems. Our findings underscore the immense potential of Big Data technology to revolutionize healthcare fraud detection. We offer a clear roadmap to effectively harness Big Data, driving meaningful change and addressing emerging challenges in the field. This review not only analyzes how the 5Vs can be leveraged to improve fraud detection systems but also identifies existing research gaps and proposes future research directions. Furthermore, we address the integration of artificial intelligence (AI) and big data applications in healthcare fraud detection, highlighting their role in enhancing operational efficiencies and improving the overall quality of healthcare services. This examination underscores the transformative potential of these technologies in reshaping fraud prevention and fostering a more trustworthy healthcare environment.

The structure of this paper is as follows. Big data for enhanced healthcare fraud detection is discussed in Section 2. Section 3 provides big data ecosystems in healthcare. The 5Vs of Big Data in healthcare fraud detection is given in Section 4. Section 5 offers an in-depth literature review, examining existing healthcare fraud detection techniques and emphasizing the critical role of big data. In Section 6, we detail the key findings in this study. Section 7 presents the detailed analysis and implications of the challenges posed by the 5Vs of Big Data. Section 8 offers the integration of AI and big data applications for healthcare fraud detection. Section 9 discusses the research directions for 5V's challenges in healthcare fraud detection. Section 10 provides the concrete suggestions for 5Vs challenges in healthcare fraud detection. Finally, Section 11 concludes the paper.

2 Big data for enhanced healthcare fraud detection

Big data analytics plays a crucial role in detecting and preventing healthcare fraud, which poses significant financial and operational challenges to the healthcare industry. The integration of advanced data analytics techniques, such as machine learning, artificial intelligence, and data mining, has enhanced the ability to identify and mitigate fraudulent activities in healthcare insurance claims. These techniques allow for real-time detection and prevention of fraud, uncovering patterns and anomalies in large datasets, and improving the overall efficiency of fraud detection systems. The following sections explore various methodologies and challenges associated with using big data for healthcare fraud detection.

2.1 Techniques in healthcare fraud detection

- *Descriptive and predictive analytics*: Descriptive analytics helps uncover patterns and anomalies in historical claims data, while predictive analytics uses statistical models and machine learning to forecast potential fraud. These techniques are essential for real-time fraud detection and prevention [16].
- *Data mining and unsupervised learning*: Association rule mining, combined with unsupervised learning techniques, is used to detect healthcare insurance fraud. This approach involves extracting frequent rules from transactions and using unsupervised classifiers to identify fraudulent activities [3].
- *Graph structure learning*: Multi-channel heterogeneous graph structure learning constructs graphs from various entities in health insurance data, using deep learning methods to detect anomalies. This method provides a promising solution for detecting complex and evolving fraudulent activities [17].
- *Bayesian belief networks*: These networks exploit the relational structure of attributes in transactions, offering better interpretability and comparable performance to traditional machine learning models [1].

2.2 Challenges in fraud detection

- *Data quality and privacy concerns*: the complexity and quality of healthcare data, along with privacy concerns, pose significant challenges to effective fraud detection. Ensuring data quality and addressing privacy issues are critical for the success of fraud detection systems [3, 16].
- *Imbalanced data and high dimensionality*: handling imbalanced datasets and high dimensionality is a

common challenge in Medicare fraud detection. Techniques like Random Undersampling (RUS) and feature selection are employed to optimize machine learning models and improve classification performance [6, 10].
- *Evolving fraud schemes*: fraud schemes continuously evolve, adapting to detection methods. This necessitates the development of more accurate and flexible analytics to keep pace with fraudsters' innovations [17].

2.3 Enhancements and future directions

- *Data-centric approaches*: enriching datasets with new features and employing data-centric machine learning workflows can significantly improve fraud detection performance. This approach encourages better data understanding and preparation techniques [18].
- *Cybersecurity integration*: combining data science with robust cybersecurity measures, such as encryption and access controls, enhances the protection of sensitive patient information and system integrity, contributing to a proactive approach in combating healthcare fraud [19].

While big data analytics offers substantial benefits in detecting healthcare fraud, it is essential to address the challenges of data quality, privacy, and evolving fraud schemes. The integration of advanced techniques, such as graph structure learning and Bayesian networks, along with a focus on data-centric approaches and cybersecurity, can further enhance fraud detection capabilities.

3 Big data ecosystems in healthcare fraud detection

Hadoop and Spark are two prominent big data processing frameworks [20] that can be leveraged for healthcare fraud detection, offering distinct advantages in handling large datasets and complex computations. Both frameworks can be utilized to enhance the efficiency and accuracy of fraud detection systems in healthcare by processing vast amounts of data and applying sophisticated machine learning algorithms.

3.1 Hadoop in healthcare fraud detection

- *MapReduce framework*: Hadoop's MapReduce framework is utilized for processing large datasets in parallel, which is beneficial for healthcare fraud detection. The MapReduce-based iterative support vector machine (MR-ISVM) classifier is an example where Hadoop's capabilities are harnessed to improve classification accuracy and reduce computational time in detecting fraudulent claims in health insurance databases. The MR-ISVM classifier achieved an accuracy of 87.73%, outperforming other SVM kernel classifiers [21].
- *Data integration and processing*: Hadoop's ability to handle diverse data sources is crucial for integrating multiple Medicare datasets, as demonstrated in studies focusing on Medicare fraud detection. By combining datasets, researchers can improve the detection performance, as seen with the Combined dataset yielding the best results in fraud detection using logistic regression [8].

3.2 Spark in healthcare fraud detection

- *Real-time processing*: Apache Spark's in-memory processing capabilities make it suitable for real-time fraud detection, which is essential in rapidly identifying and responding to fraudulent activities. Spark's ability to process data faster than Hadoop's MapReduce can be advantageous in scenarios requiring immediate action.
- *Machine learning integration*: Spark's MLlib library facilitates the integration of machine learning algorithms for fraud detection. This allows for the application of various models, such as random forests and neural networks, which have been shown to be effective in detecting healthcare fraud with high accuracy and precision [22].

3.3 Advanced techniques in fraud detection

- *Graph-based methods*: Recent advancements in fraud detection involve the use of graph-based methods, such as Multi-channel Heterogeneous Graph Structured Learning (MHGSL) and Attributed Heterogeneous Information Networks (AHIN). These methods utilize graph neural networks to capture complex relationships and interactions within healthcare data, leading to improved fraud detection accuracy [17, 23].
- *Attention mechanisms*: The use of hierarchical attention mechanisms in models like MHAMFD enhances the interpretability and accuracy of fraud detection by focusing on significant behavioral relationships between patients' visits [23].

While Hadoop and Spark provide robust frameworks for processing large datasets, the integration of advanced machine learning techniques, such as graph-based models and attention mechanisms, further enhances the capability of fraud detection systems. These methods address the complexity and evolving nature of healthcare fraud, offering promising solutions for improving the fairness and sustainability of health insurance systems. However, the

choice between Hadoop and Spark may depend on specific requirements, such as the need for real-time processing or batch processing capabilities.

4 5Vs of big data in enhancing healthcare fraud detection

The concept of the 5Vs of Big Data (Volume, Variety, Velocity, Veracity, and Value) plays a crucial role in enhancing healthcare fraud detection. These dimensions help in managing and analyzing large datasets to identify fraudulent activities effectively. By leveraging these aspects, healthcare systems can improve the accuracy and efficiency of fraud detection mechanisms, ultimately reducing financial losses and improving patient care. Below, each of the 5Vs is explored in the context of healthcare fraud detection.

Volume

The vast amount of data generated by healthcare systems, including electronic medical records and insurance claims, provides a rich source for detecting fraud. For instance, the use of large-scale Medicare datasets has been shown to improve fraud classification performance by enriching data with new features.

Variety

Healthcare data comes in various forms, such as structured data from claims and unstructured data from medical records. Integrating diverse data types enhances the detection of complex fraud patterns. For example, combining different Medicare datasets has been effective in improving fraud detection performance.

Velocity

Real-time data processing is essential for timely fraud detection. Advanced techniques like machine learning and artificial intelligence enable the analysis of data as it is generated, allowing for immediate identification of anomalies and potential fraud.

Veracity

Ensuring data accuracy and reliability is critical for effective fraud detection. Data-centric approaches that focus on data quality and preparation, such as improved data labeling and feature selection, have been shown to enhance the reliability of fraud detection models.

Value

The ultimate goal of using Big Data in fraud detection is to derive actionable insights that lead to cost savings and improved healthcare services. By reducing class imbalance and improving model performance, techniques like FraudNET and Random Undersampling have demonstrated significant improvements in fraud detection accuracy.

While the 5Vs provide a robust framework for enhancing healthcare fraud detection, challenges such as data privacy concerns and the need for continuous model updates remain. Addressing these issues through regulatory frameworks and collaborative efforts is essential for maximizing the benefits of Big Data in this domain.

5 Literature review and experimental results

The paper [24] presents an innovative approach to healthcare fraud detection by evaluating the trustworthiness of doctors and identifying copying prescription behavior. By combining graph mining with frequent pattern recognition, the proposed method constructs a Rational Treatment Model (RTM) to distinguish fraudulent records from normal ones. The approach leverages the large volume and variety of healthcare data, ensuring high veracity and delivering significant value by improving fraud detection accuracy. Although the paper doesn't specify the use of Big Data platforms, the techniques discussed are well-suited for deployment in environments like Apache Spark or Hadoop, enabling efficient processing of vast healthcare datasets. Table 1 provides a detailed overview of the 5Vs considerations for this research work.

The paper [25] presents an in-depth review of techniques for detecting health care fraud, with a specific focus on the importance of utilising big data analytics. The article examines the attributes of big data specifically in relation to detecting health care fraud. It emphasises the difficulties and remedies involved in handling and analysing extensive, varied, and quickly produced health care data. The study highlights the significance of utilising new analytical techniques and big data platforms, like Hadoop, to enhance the efficiency and efficacy of fraud detection systems. The ultimate goal is to decrease costs and enhance the quality of healthcare. Table 2 provides a detailed overview of the 5Vs considerations for this research work.

The paper [26] investigates the effect of class imbalance on Medicare fraud detection using big data. By analyzing the 2012–2015 Medicare Provider Utilization and Payment Data along with the LEIE database, the researchers demonstrate that a 90:10 class distribution provides the best results for fraud detection, debunking the common use of a 50:50 distribution. The findings highlight the importance of proper data sampling in big data environments to improve model performance and suggest that addressing class imbalance is crucial for effective fraud detection. Table 3 provides a detailed overview of the 5Vs considerations for this research work.

The paper [8] explores Medicare fraud detection by leveraging Big Data techniques on datasets released by the Centers for Medicare and Medicaid Services (CMS). It combines multiple datasets (Part B, Part D, DMEPOS) to enhance fraud detection capabilities, using machine

Table 1 Key aspects of contribution in [24]

Points	Observations
Volume	The paper addresses the Volume aspect by utilizing a substantial healthcare dataset to detect fraudulent activities. The large volume of data allows for a comprehensive analysis and a robust fraud detection model.
Variety	The variety characteristic is not highlighted and the data includes attributes such as patient records, doctor information, treatment details, and timestamps.
Velocity	Velocity is considered in the approach's ability to process and analyze healthcare data rapidly. The paper doesn't explicitly mention real-time processing, but the methods used suggest an efficient handling of data, which is crucial for timely fraud detection.
Veracity	The Veracity of the data is ensured by focusing on the trustworthiness of doctors and identifying copying prescription behavior. The approach aims to filter out fraudulent records accurately, thereby maintaining high data integrity and reliability.
Value	The Value derived from this research is significant, as it proposes a novel method for detecting healthcare fraud, which can save costs and protect the interests of insured individuals. The detection model adds value by improving the accuracy and efficiency of identifying fraudulent activities in healthcare.
Environmental setup	The paper doesn't explicitly state the use of specific Big Data environments like Spark or Hadoop. However, the methodologies discussed, such as graph mining and frequent pattern recognition, could be effectively implemented using these frameworks.

Table 2 Key aspects of contribution in [25]

Points	Observations
Volume	The paper highlights the immense volume of data generated by the health care industry, driven by record-keeping, patient care, and electronic claims processing. The sheer size of these data sets poses significant challenges for traditional data management techniques, necessitating the use of big data analytics to handle and process this information effectively.
Variety	Health care data comes from various sources, including insurance claims, governmental health department data, clinical-instance data, general practitioners' data, and private insurance companies' data. This data is heterogeneous, encompassing different formats and types, which include billing records, patient records, drug prescriptions, and more. The diversity of data sources and types underscores the need for sophisticated tools to integrate and analyze these varied data sets.
Velocity	The paper discusses the rapid generation and need for real-time processing of health care data. Modern fraud detection systems must handle continuous streams of data from electronic claims and other health care transactions. The high velocity of data necessitates the use of big data technologies capable of processing and analyzing data in real-time or near real-time to detect fraudulent activities promptly.
Veracity	Ensuring the accuracy and trustworthiness of health care data is crucial for effective fraud detection. The paper acknowledges the challenges posed by the inherent uncertainties and ambiguities in health care data. High veracity is essential to minimize false positives and negatives in fraud detection, requiring robust data cleaning and validation techniques.
Value	The ultimate goal of applying big data analytics in health care fraud detection is to derive value by reducing fraud, waste, and abuse. By effectively identifying fraudulent patterns and behaviors, these systems can save significant costs for health care systems and improve the quality of care. The paper emphasizes that big data analytics can enhance the efficiency of fraud detection and provide high-quality insights to health care systems.
Environmental setup	The paper references the use of big data platforms like Hadoop for handling large-scale health care data. Hadoop's distributed storage and processing capabilities make it suitable for managing the massive volumes of diverse health care data. Additionally, methods such as Naive Bayes, decision trees, and clustering have been applied to big data analysis on the Hadoop platform, demonstrating its utility in health care fraud detection.

learning models like Logistic Regression, Random Forest, and Gradient Tree Boosting. The study highlights the importance of using comprehensive data for effective fraud detection, achieving the best results with a combined dataset. Utilizing Apache Spark on a Hadoop YARN cluster, it efficiently processes large volumes of data, underscoring the potential for Big Data analytics to improve healthcare systems by identifying fraudulent activities and reducing financial losses. Table 4 provides a detailed overview of the 5Vs considerations for this research work.

The paper [27] presents a framework for fraud detection in government-supported national healthcare programs, focusing on detecting anomalies in insurance claims using

Table 3 Key aspects of contribution in [26]

Points	Observations
Volume	The study deals with substantial amounts of data, including multiple years of Medicare claims across several datasets (Part B, Part D, DMEPOS).
Variety	Combines diverse datasets that cover different aspects of Medicare claims, including physician services, prescription drugs, and medical equipment referrals. This variety provides a more holistic approach to fraud detection.
Velocity	Although not explicitly discussed, the datasets are updated annually, indicating a regular influx of new data that needs timely processing.
Veracity	The data from CMS is considered reliable and accurate, with transparent quality controls, which is crucial for trustworthy fraud detection outcomes.
Value	By integrating and analyzing these datasets, the study enhances the capability to detect fraudulent activities, potentially saving significant amounts of money and improving healthcare system efficiency.
Environmental setup	The study uses Apache Spark on a Hadoop YARN cluster to efficiently handle and process the large datasets. Spark is well-suited for Big Data processing due to its speed and ability to perform complex analytics, which is essential for the large-scale data involved in this research.

Table 4 Key aspects of contribution in [8]

Points	Observations
Volume	The paper deals with a substantial dataset, consisting of millions of Medicare records. This showcases the ability to process and analyze large volumes of data effectively.
Variety	The study integrates different data sources such as Medicare claims and the LEIE fraud labels. This involves handling various data types and structures, from numerical values to categorical labels.
Velocity	While the paper focuses on historical data, it doesn't specifically address the speed of data processing or real-time analytics. The emphasis is on improving fraud detection models using existing data.
Veracity	Ensuring data quality is a key focus. The authors carefully map fraud labels and filter the data to maintain accuracy, using known exclusion rules to validate the data's integrity.
Value	The research provides actionable insights by identifying the optimal class distribution for fraud detection, enhancing the effectiveness of Medicare fraud detection models.
Environmental setup	The paper does not mention the use of Hadoop or Spark. Instead, it centers on using Random Forest models and random undersampling techniques to address class imbalance in the dataset. This suggests a focus on statistical and machine learning methods rather than distributed computing frameworks.

time series analysis and dynamic rating scores. It addresses the Big Data characteristics by handling large datasets from diverse sources, ensuring real-time processing, maintaining data integrity, and providing substantial value in terms of fraud prevention. The architecture of the proposed system utilises OLAP tools and machine learning methods, allowing for seamless connection with Big Data environments such as Spark or Hadoop. This integration enhances the system's processing capabilities and efficiency. Table 5 provides a detailed overview of the 5Vs considerations for this research work.

The authors [28] detail a methodological approach adopted by the Hellenic National Organization for the Provision of Health Services (EOPYY) to leverage Big Data for fraud detection in healthcare claims. By analyzing a vast dataset of over 4 million prescriptions from 2018, using advanced outlier detection algorithms like LOCI and ABOD, the study identifies potential fraudulent activities. The results reveal seven significant outliers, two of which have already been confirmed as fraud cases by EOPYY's auditing services. This approach underscores the pivotal role of Big Data in enhancing the accuracy and efficiency of fraud detection, ultimately contributing to reduced healthcare expenditures and improved service quality. Table 6 provides a detailed overview of the 5Vs considerations for this research work.

The paper [29] proposes a novel method combining MapReduce and association rule mining to detect fraud in healthcare insurance. It effectively processes 1.5 million medical records to identify abnormal patterns. The method's integration into a Hadoop-based environment demonstrates its practical application and scalability. This approach significantly enhances the accuracy of fraud detection, providing valuable insights and decision support for managing medical insurance funds. Table 7 provides a

Table 5 Key aspects of contribution in [27]

Points	Observations
Volume	The paper deals with a significant volume of data, as evidenced by the dataset used, which consists of 63,000 records of employee transactions. This large volume is typical in healthcare fraud detection, where numerous transactions and claims need to be analyzed.
Variety	The data used in the paper comes from multiple sources within a hospital environment, including departments like admission, cardiology, neurology, and reception. This variety includes different data types such as patient demographics, service records, and financial transactions, which are crucial for identifying fraudulent activities.
Velocity	The framework proposed in the paper emphasizes real-time fraud detection and monitoring. The use of time series traces and dynamic rating scores for patients, providers, and services suggests a focus on processing and analyzing data at high speed to detect anomalies as they occur.
Veracity	The paper addresses data veracity by implementing data cleaning and reduction processes. Anonymization of patient data ensures privacy protection, enhancing the reliability and trustworthiness of the data used in the fraud detection system.
Value	The proposed framework aims to provide significant value by detecting and isolating fraudulent activities in healthcare transactions. By doing so, it helps in reducing financial losses due to fraud, waste, and abuse, thereby increasing the efficiency and quality of healthcare services.
Environmental setup	The paper mentions the use of OLAP (Online Analytical Processing) tools and machine learning algorithms for analyzing multidimensional data. While specific Big Data technologies like Spark or Hadoop are not explicitly mentioned, the architecture described suggests a data warehousing approach that could be integrated with Big Data platforms for enhanced processing capabilities.

Table 6 Key aspects of contribution in [28]

Points	Observations
Volume	The paper deals with a massive amount of prescription data amounting to over 4,000,000 entries from 2018, stored in a database of approximately 5 TB with a monthly increase rate of 140 GB.
Variety	The data includes various types of information such as prescription dates, prescribers' identities and specialties, patients' social security numbers, product names, categories, providers' identities, and locations.
Velocity	The data is continually growing, reflecting the ongoing prescription activities. The paper does not explicitly mention real-time processing, but the increase rate indicates a dynamic and continuously updating dataset.
Veracity	The data undergoes anonymization to comply with GDPR, ensuring the reliability and ethical handling of sensitive information. The analysis also focuses on extracting accurate insights by identifying and investigating outliers.
Value	The primary value derived from this Big Data analysis is the detection of potential fraud cases in healthcare claims, which can significantly reduce healthcare costs and improve service quality.
Environmental setup	The paper does not explicitly mention the specific Big Data technologies or platforms used (such as Spark or Hadoop). However, the methodologies and algorithms applied (Local Correlation Integral (LOCI) and Angle-based outlier detection (ABOD)) suggest a setup capable of handling large datasets and performing complex calculations, which are typical features of Big Data environments like Spark or Hadoop.

detailed overview of the 5Vs considerations for this research work.

The paper [30] demonstrates a methodological approach to detecting financial and medical fraud using Big Data technologies. By analyzing 4 million prescription entries from the EOPYY database, the study successfully identifies outliers using the Local Correlation Integral (LOCI) algorithm. The research highlights the importance of the Big Data 5Vs in processing and analyzing healthcare data. Although the specific Big Data environment is not mentioned, the techniques used are indicative of advanced data processing capabilities. The findings offer significant value in enhancing fraud detection processes and ensuring the efficient use of healthcare resources. Table 8 provides a detailed overview of the 5Vs considerations for this research work.

The paper [31] presents a robust framework for detecting fraud in Nigeria's National Health Insurance Scheme (NHIS) using big data analytics. By leveraging tools such as Apache Hive, HDFS, and MapReduce, the study addresses the key characteristics of big data. The proposed system aims to enhance the integrity and efficiency of healthcare administration in Nigeria by identifying fraudulent activities and improving resource allocation. Table 9 provides a detailed overview of the 5Vs considerations for this research work.

Table 7 Key aspects of contribution in [29]

Points	Observations
Volume	The paper addresses the challenge of analyzing 1.5 million medical claim records to detect abnormal behaviors. The large scale of data showcases the Volume characteristic of Big Data.
Variety	The dataset includes diverse types of medical insurance records, which encompass different patient behaviors, medical treatments, and reimbursement activities. This diversity reflects the Variety in Big Data.
Velocity	The proposed method leverages the MapReduce distributed computing model to process data efficiently, ensuring that the detection of fraud and anomalies can happen in a timely manner. This highlights the Velocity aspect of Big Data.
Veracity	By mining frequent patterns and employing association rules, the method aims to improve the accuracy and reliability of fraud detection. This approach ensures the trustworthiness of the results, addressing the Veracity characteristic of Big Data.
Value	The ultimate goal of the proposed method is to reduce healthcare insurance fraud and improve the management of medical insurance funds. By doing so, it adds significant Value to the healthcare system by safeguarding public funds and enhancing medical quality.
Environmental setup	The paper utilizes a Hadoop environment to set up the distributed anomaly detection model. Hadoop's MapReduce framework is employed for processing the large dataset and mining the required patterns. This setup ensures that the handling of Big Data is both efficient and scalable.

Table 8 Key aspects of contribution in [30]

Points	Observations
Volume	The paper deals with a substantial Volume of data, specifically 5 TB of healthcare provision and claims data from Greek citizens, increasing by 140 GB per month. The study focused on analyzing 4 million prescription entries from 2018.
Variety	The dataset includes a diverse range of information such as prescription dates, prescribers' details, patients' social security numbers, product categories (e.g., diabetes management supplies, catheters, medical pads, special nutrition products), and provider locations. This wide range of data types highlights the Variety in Big Data.
Velocity	The data is continually growing at a significant rate (140 GB per month), requiring timely and efficient processing to detect fraud. The use of the Local Correlation Integral (LOCI) algorithm for outlier detection showcases the importance of Velocity in handling and analyzing data swiftly.
Veracity	The study applies anonymization techniques (SHA-1 algorithm) to ensure data privacy and comply with GDPR regulations, thereby enhancing the Veracity or trustworthiness of the data. The LOCI algorithm's precise outlier detection further supports the reliability of the analysis.
Value	The main Value derived from this study is the ability to detect healthcare fraud, which is crucial for controlling medical expenses and improving the integrity of the healthcare system. Identifying potential fraud cases helps in reducing unnecessary costs and ensuring that resources are appropriately utilized.
Environmental setup	The paper does not explicitly mention the use of specific Big Data environments like Spark or Hadoop. However, the application of the LOCI algorithm for analyzing large datasets suggests the use of advanced data processing techniques that could be implemented in environments capable of handling big data workloads.

The paper [32] highlights the critical role of maximum tree depth in enhancing the performance of ensemble learning algorithms on large-scale, high-cardinality, and imbalanced healthcare datasets. By leveraging GPU-based implementations, the research significantly reduces processing time and demonstrates substantial improvements in classification accuracy. The findings offer valuable guidance for future research and practical applications in healthcare fraud detection, emphasizing the importance of optimizing hyperparameters to achieve better outcomes. Table 10 provides a detailed overview of the 5Vs considerations for this research work.

This study [33] introduces an innovative approach to detecting medical insurance fraud by leveraging the Isolation Forest algorithm within the context of Big Data. The study improves the accuracy and effectiveness of anomaly detection by combining and integrating several data sources into a comprehensive indicator system. The proposed methodology effectively tackles the difficulties associated with managing substantial amounts of diverse data, while also enhancing the dependability and velocity of identifying fraudulent activity. This greatly enhances the protection of the integrity of medical insurance systems and the well-being of policyholders. Table 11 provides a detailed overview of the 5Vs considerations for this research work.

This paper [34] presents a comprehensive analysis of healthcare fraud detection using advanced machine

Table 9 Key aspects of contribution in [31]

Points	Observations
Volume	The paper deals with large datasets from Nigeria's National Health Insurance Scheme (NHIS), including enrolment, referral, and claim data. The data volume is significant, as it encompasses records from numerous healthcare facilities and Health Maintenance Organizations (HMOs) across Nigeria.
Variety	The study involves diverse data types, including structured data (e.g., enrolment records, claims data) and unstructured data (e.g., referral notes). This variety necessitates robust data processing tools capable of handling different data formats and structures.
Velocity	The data used in the study is both batch and streaming data, indicating the need for real-time data processing capabilities to quickly identify and respond to fraudulent activities. The velocity aspect is crucial for timely fraud detection and intervention.
Veracity	The paper addresses the challenge of data quality and trustworthiness. Given the high stakes of healthcare fraud, ensuring the accuracy and reliability of data is paramount. Methods to detect and manage false or misleading claims are a focal point of the study.
Value	The ultimate goal of the research is to derive valuable insights that can improve the efficiency of healthcare administration and reduce fraud. By leveraging big data analytics, the study aims to enhance the integrity of the NHIS and ensure better resource allocation and service delivery.
Environmental setup	The study employs a big data processing environment consisting of, (1) Apache Derby DB: used for database management, (2) Hadoop Distributed File System (HDFS): for storing large datasets across multiple nodes, (3) Apache MapReduce: for processing large data sets in parallel, and (4) Apache Hive: utilized for data querying and analysis, allowing the researchers to manage and analyze large datasets efficiently.

Table 10 Key aspects of contribution in [32]

Points	Observations
Volume	The study utilizes a massive dataset from the Centers for Medicare and Medicaid Services (CMS) Medicare Provider Utilization and Payment, Part D data. This dataset spans from 2013 to 2019 and contains approximately 175 million records, making it one of the largest datasets used for healthcare fraud detection research.
Variety	The dataset includes high-cardinality categorical features, with some features having hundreds or thousands of possible values.
Velocity	The research leverages GPU-based implementations of the Random Forest algorithm to handle the high computational demands of processing such a large dataset. The GPU implementation dramatically reduces the processing time, completing experiments in less than a day compared to several days for the CPU-based version.
Veracity	The paper addresses the issue of class imbalance, which is common in fraud detection datasets. By employing ensemble learning methods like XGBoost and Random Forest, the study aims to improve the reliability and accuracy of fraud detection despite the skewed distribution of fraudulent versus non-fraudulent records.
Value	The research demonstrates that increasing the maximum tree depth in ensemble learning algorithms significantly enhances their performance. For instance, XGBoost's AUC score improves from 0.75633 to 0.97273, and Random Forest's AUC score increases from 0.80821 to 0.96996. These findings provide valuable insights for future researchers and practitioners in healthcare fraud detection, potentially leading to more effective fraud prevention and better allocation of healthcare resources.
Environmental setup	The study utilizes GPU-based implementations of the Random Forest algorithm, specifically the CuML implementation from the RAPIDS AI suite. This setup accelerates the processing of large datasets and makes the computationally intensive task of hyperparameter tuning more feasible.

learning techniques on extensive and varied datasets. By addressing the critical issue of class imbalance with SMOTE and employing robust evaluation metrics, the study underscores the importance of accurate and reliable predictive models in mitigating healthcare fraud. The findings contribute valuable insights into the detection of fraudulent activities, highlighting the need for efficient data processing environments and the potential for significant cost savings and improved trust in the healthcare system. Table 12 provides a detailed overview of the 5Vs considerations for this research work.

This paper [35] focuses on detecting fraud in Medicare Part B, which includes activities like patient abuse, neglect, and billing for services not provided by excluded providers. The dataset used for fraud detection is highly imbalanced, with very few instances of fraud compared to non-

Table 11 Key aspects of contribution in [33]

Points	Observations
Volume	The paper deals with extensive datasets from the Hospital Information System (HIS) of a hospital in City K. These datasets include, (1) Patient Information Table: 58,017 medical records with 87 fields, (2) Expense Details Table: 289,799 records with 20 fields, and (3) Medical Advice Categories Table: 29,434 records with 159 fields.
Variety	The datasets encompass a variety of data types and sources, including (1) Patient demographics and medical history, (2) Detailed expense information, and (3) Prescriptions and medical advice records. The study integrates these heterogeneous data sources to create a comprehensive indicator system that provides a detailed view of individual patient behaviors.
Velocity	While the paper does not explicitly mention the speed of data processing, the use of the Isolation Forest algorithm suggests a focus on efficient anomaly detection. The algorithm's design, which employs binary trees for partitioning data, inherently reduces time complexity, thus facilitating faster detection of anomalies.
Veracity	The study addresses data quality and labeling challenges by (1) Aggregating data to generate comprehensive features, (2) Mitigating issues related to diverse and heterogeneous data sources, and (3) Removing noise points, such as negative monthly prescription drug counts. This approach enhances the reliability of the data used for anomaly detection.
Value	The primary value derived from this study lies in its ability to (1) Enhance audit efficiency and accuracy in detecting medical insurance fraud, (2) Improve the generalization and effectiveness of the anomaly detection model, and (3) Support regulatory authorities in recovering unlawfully obtained medical insurance funds and enhancing early warning systems.
Environmental setup	The paper does not specify a particular Big Data environment such as Spark or Hadoop. However, given the nature of the data processing and the use of Python for model implementation, it is likely that the study could benefit from environments that support large-scale data processing and machine learning.

Table 12 Key aspects of contribution in [34]

Points	Observations
Volume	The study leverages comprehensive datasets encompassing Inpatient claims, Outpatient claims, and Beneficiary information, indicating a large volume of data. This extensive data collection is crucial for the robust detection of fraudulent activities in healthcare.
Variety	The datasets include diverse types of data such as claims data from various providers, physicians, and beneficiaries. The variety of data sources and types enriches the analysis, allowing for more accurate and nuanced detection of fraud patterns.
Velocity	The paper does not explicitly mention the speed at which data is processed. However, the use of advanced machine learning models like Random Forest and Logistic Regression suggests the need for efficient and timely data processing to detect fraud in real-time or near real-time scenarios.
Veracity	The study addresses data quality and reliability by employing techniques like the Synthetic Minority Over-sampling Technique (SMOTE) to handle class imbalance, thereby improving the robustness and accuracy of the predictions. This indicates a focus on ensuring that the data used is trustworthy and the insights derived are reliable.
Value	The primary value derived from this study is the enhanced capability to predict and detect healthcare fraud, which can significantly reduce healthcare costs and improve the integrity of the healthcare system. By identifying fraudulent providers, the study aims to mitigate financial losses and ensure more efficient allocation of resources within the healthcare sector.
Environmental setup	The paper does not explicitly mention the use of specific Big Data environments like Spark or Hadoop. However, given the nature of the data and the analytical techniques employed, it is likely that a robust data processing environment capable of handling large-scale data and complex computations, such as Spark or Hadoop, would be beneficial.

fraudulent cases. To address this imbalance, the researchers created seven different class distributions and tested six machine learning methods to find the best approach for detecting fraud. The Random Forest method with a 90:10 class distribution showed the best performance, achieving an AUC of 0.87302. The study suggests that retaining more information from the majority class leads to better fraud detection performance across most machine learning models. Table 13 provides a detailed overview of the 5Vs considerations for this research work.

This paper [36] introduces a new system called Provider Fraud Anomaly Detection System (PFADS) that uses big data and deep learning to detect fraud in healthcare insurance, focusing on improving preprocessing and classification phases to identify fraud early on the preprocessing phase uses a Relative Risk-based MapReduce framework to organize relationships between diseases, patients, and claiming variables, which helps in better data management and analysis. The classification phase utilises a Recurrent Neural Network (RNN) with hyperparameter

Table 13 Key aspects of contribution in [35]

Points	Observations
Volume	The study deals with a massive dataset of Medicare Part B claims, which includes a large number of records to identify fraudulent activities by providers.
Variety	The data includes different types of information such as billing records, patient details, and service codes, which are used to detect various forms of fraud like patient abuse and billing for services not rendered.
Velocity	The data processing involves handling and analyzing data quickly to detect fraud in a timely manner, ensuring that fraudulent activities are identified as soon as possible.
Veracity	The authors focus on ensuring the accuracy and reliability of the data by mapping fraud labels with known fraudulent providers, which helps in maintaining the trustworthiness of the datase.
Value	The ultimate goal of the study is to provide valuable insights and effective methods for detecting Medicare fraud, which can save significant amounts of money and improve the integrity of the healthcare system.
Environmental setup	While the paper discusses the importance of handling large volumes of data and ensuring data accuracy, they did not mention the use of Big Data processing ecosystems like Hadoop or Spark.

optimisation to improve the identification of anomalies, taking advantage of the RNN's capacity to remember past and current states of the network. The PFADS framework has gone through testing and validation using public datasets, demonstrating superior performance in accuracy, precision, recall, and computational time while compared to alternative approaches. Table 14 provides a detailed overview of the 5Vs considerations for this research work.

This paper [21] presents a novel approach to fraud detection in healthcare insurance using a MapReduce-based iterative support vector machine (MR-ISVM). The study effectively improves the accuracy and efficiency of detecting fraudulent claims by utilising Big Data principles. The implementation of the MapReduce architecture showcases the capacity for scalable and dependable data processing solutions in the healthcare insurance sector, resulting in substantial cost reductions and enhanced decision-making. Table 15 provides a detailed overview of the 5Vs considerations for this research work.

6 Key findings

Table 16 summarizes the key aspects of contributions from various studies (Table 1 to Table 14) regarding the impact of big data characteristics on healthcare fraud detection. The table highlights how different dimensions of big data (volume, variety, velocity, veracity, and value) affect healthcare fraud detection models and their outcomes. This synthesis offers a thorough examination of the empirical evidence that substantiates the incorporation of big data in healthcare fraud detection, emphasising the advantages and obstacles that are linked to each characteristic. The key findings include:

- Volume: Large datasets, including up to 175 million records, demonstrate the necessity of managing vast amounts of healthcare data for effective fraud detection. Handling extensive data volumes is crucial for uncovering hidden patterns and anomalies, leading to more comprehensive fraud detection.

Table 14 Key aspects of contribution in [36]

Points	Observations
Volume	The paper deals with large datasets by using a Relative Risk-based MapReduce framework to organize relationships between diseases, patients, and claiming variables, which helps in managing and analyzing vast amounts of data efficiently.
Variety	The system integrates various types of data, including patient records, disease information, and claim variables, to create a comprehensive dataset that can be used for detecting fraud in healthcare insurance.
Velocity	The proposed framework processes data quickly, as evidenced by the computational time of 92.30 seconds, ensuring timely detection of fraudulent activities in healthcare insurance claims.
Veracity	The use of a Recurrent Neural Network (RNN) with hyperparameter optimization helps in improving the accuracy and reliability of the fraud detection system, reducing false positives and enhancing the system's trustworthiness.
Value	By detecting fraud early and accurately, the system provides significant value by potentially saving public funds and improving the overall quality of healthcare services.
Environmental setup	There is no specific mention of using Spark in the paper, but the MapReduce framework used for data processing is a core component of Hadoop, indicating that Hadoop is utilized for handling big data tasks.

Table 15 Key aspects of contribution in [21]

Points	Observations
Volume	The research deals with extensive datasets from healthcare insurance claims. This large volume of data is crucial for detecting patterns and anomalies that indicate fraud.
Variety	The data includes different types, such as structured information like claim IDs, disease codes, and treatment details, which is not an indication of variety consideration.
Velocity	The system is designed to process data rapidly, achieving faster detection of fraudulent activities. This is essential in preventing further losses and ensuring timely responses.
Veracity	In this paper, ensuring data accuracy and reliability is a priority. The model aims to minimize false positives and negatives, which enhances trust in the detection outcomes.
Value	The proposed method adds value by effectively identifying fraudulent claims, which can lead to significant cost savings and improved resource allocation in the healthcare insurance industry.
Environmental setup	The paper explicitly uses the MapReduce framework, a core component of the Hadoop ecosystem, to manage and process large datasets.

- Variety: Incorporating diverse data types such as billing records, patient demographics, and service codes enriches the detection process, allowing for identification of varied fraudulent activities. The integration of structured and unstructured data sources provides a holistic view, enhancing the ability to detect complex fraud patterns.
- Velocity: Real-time or near real-time processing is essential to promptly identify and mitigate fraudulent activities, reducing potential financial losses. Techniques like MapReduce and GPU-based processing significantly enhance data handling speed and efficiency, supporting timely fraud detection.
- Veracity: Ensuring data accuracy and reliability through methods such as data anonymization, noise reduction, and mapping fraud labels with known cases is critical for trustworthy outcomes. Addressing class imbalance with techniques like SMOTE improves prediction accuracy, reducing false positives and enhancing system credibility.
- Value: Effective fraud detection leads to substantial cost savings, preserving public funds and strengthening the integrity of healthcare systems. Improved audit processes and resource allocation result in better service delivery, enhancing patient trust and healthcare quality.

Table 17 summarizes the extent to which the 12 reviewed papers incorporate the 5 V's (Volume, Variety, Velocity, Veracity, and Value) and the usage of Spark/Hadoop frameworks. A "cross" (lightred×) indicates that the paper did not consider the respective characteristic or framework, while a "check" (lightgreen✔) means it was considered.

7 Detailed analysis and implications

This section delves into the detailed analysis and implications of the challenges posed by the 5Vs of Big Data - Volume, Variety, Velocity, Veracity and Value- in the context of healthcare fraud detection. The analysis and implications are supported by Figure 1, which visualizes these challenges and implications.

7.1 Volume

The integration of Big Data into healthcare fraud detection has the potential to transform how fraudulent activities are identified and prevented. However, the Volume aspect of Big Data presents substantial challenges that must be addressed to fully exploit its advantages. The immense scale of healthcare data, encompassing billing records, patient information, and medical claims, requires robust computational frameworks for effective analysis. The significant challenge lies in efficiently processing and analyzing terabytes of data to uncover fraudulent patterns in real-time. This requires not only substantial computational resources but also sophisticated algorithms capable of handling such voluminous data. An analysis of the reviewed studies (Tables 1 to 14) reveals several critical shortcomings:

1. *Inadequate Volume Handling:* A majority of the studies (Tables 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14) do not adequately address the Volume dimension, indicating a significant gap in managing large-scale datasets effectively.

2. *Limited Utilization of Big Data Ecosystems:* As reflected in Table 17, only a few studies (3, 7, 9, 14) utilize advanced data processing frameworks like Spark or Hadoop. These tools are essential for scaling processing capabilities and enabling real-time data analysis.

Table 16 Overview of key contributions from research papers applying big data and advanced analytics to healthcare fraud detection

Tables	Volume	Variety	Velocity	Veracity	Value
1	Utilizes substantial healthcare datasets for robust fraud detection.	Analyzes diverse data types like patient records and timestamps.	Suggests efficient handling of data for timely detection.	Ensures trustworthiness by focusing on doctor reliability.	Proposes a novel method for improved fraud detection.
2	Highlights immense volume from health care industry data.	Data from various sources like insurance claims and clinical data.	Discusses need for real-time processing of health care data.	Emphasizes accuracy to minimize false positives and negatives.	Reduces fraud, waste, and abuse; enhances care quality.
3	Substantial datasets from Medicare claims across multiple years.	Combines datasets like Part B, Part D, DMEPOS for holistic fraud detection.	Annual updates indicate regular data influx.	Reliable CMS data with transparent quality controls.	Enhances fraud detection, saving money, improving efficiency.
4	Millions of Medicare records processed effectively.	Integrates Medicare claims and LEIE fraud labels.	Focuses on historical data, improving detection models.	Maps fraud labels accurately, ensuring data integrity.	Identifies optimal class distribution for detection.
5	Deals with 63,000 records of employee transactions.	Data from multiple hospital departments.	Emphasizes real-time detection using time series.	Ensures privacy through anonymization.	Detects and isolates fraudulent activities to reduce losses.
6	4 million prescriptions, 5 TB database.	Includes prescription data, identities, and locations.	Continual data growth, indicating dynamic updates.	Anonymizes data for GDPR compliance.	Detects potential fraud, reducing costs and improving quality.
7	Analyzes 1.5 million medical records.	Includes diverse medical insurance records.	Uses MapReduce for timely fraud detection.	Ensures trustworthiness through pattern mining.	Reduces insurance fraud, improving fund management.
8	5 TB of healthcare data, 4 million entries.	Diverse data including prescriptions and categories.	Continual growth at 140 GB/month.	Uses SHA-1 for data privacy.	Detects fraud, reducing costs and improving integrity.
9	Large datasets from Nigeria's NHIS.	Structured and unstructured data types.	Batch and streaming data for real-time detection.	Ensures data accuracy and reliability.	Improves healthcare administration efficiency.
10	175 million records from CMS.	High-cardinality categorical features.	GPU-based processing for reduced time.	Addresses class imbalance for accuracy.	Improves fraud detection with optimized algorithms.
11	Extensive datasets from HIS in City K.	Diverse data types and sources, including patient and expense info.	Isolation Forest for efficient anomaly detection.	Enhances reliability by aggregating data and removing noise.	Improves audit efficiency and fraud detection accuracy.
12	Comprehensive datasets, including claims and beneficiary info.	Diverse data from various providers and beneficiaries.	Advanced ML models for efficient processing.	Uses SMOTE to handle class imbalance for robust predictions.	Enhances fraud detection, reducing costs and improving trust.
13	Massive dataset of Medicare Part B claims.	Includes billing records, patient details, and service codes.	Quick data analysis for timely fraud detection.	Maps fraud labels with known fraudulent providers.	Provides effective methods for detecting Medicare fraud.
14	Utilizes MapReduce for large datasets.	Integrates patient records, disease info, and claim variables.	Processes data in 92.30 seconds for timely detection.	RNN with hyperparameter optimization for accuracy.	Saves public funds and improves healthcare quality.
15	Extensive datasets for fraud detection.	Structured data like claim IDs and codes.	Rapid processing for timely fraud detection.	Ensures accuracy and minimizes errors.	Significant cost savings and resource efficiency.

3. *Scalability Issues:* Traditional data processing techniques used in many studies are not scalable, limiting their effectiveness as data volumes continue to grow exponentially.

7.1.1 Implications for future research: volume

Addressing these challenges is imperative for advancing fraud detection systems. Future research should focus on:

Table 17 Application of Big Data 5Vs in the reviewed studies on Healthcare Fraud Detection

Table	Volume	Variety	Velocity	Veracity	Value	Big Data Ecosystem
1	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
2	(Lightred×)	(Lightred×)	(Lightred×)	(Lightred×)	(Lightred×)	(Lightred×)
3	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightgreen✔)
4	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
5	(Lightred×)	(Lightgreen✔)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
6	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
7	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightgreen✔)
8	(Lightred×)	(Lightgreen✔)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
9	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightgreen✔)
10	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
11	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
12	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
13	(Lightred×)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightred×)
14	(Lightgreen✔)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightgreen✔)
15	(Lightgreen✔)	(Lightred×)	(Lightred×)	(Lightgreen✔)	(Lightgreen✔)	(Lightgreen✔)

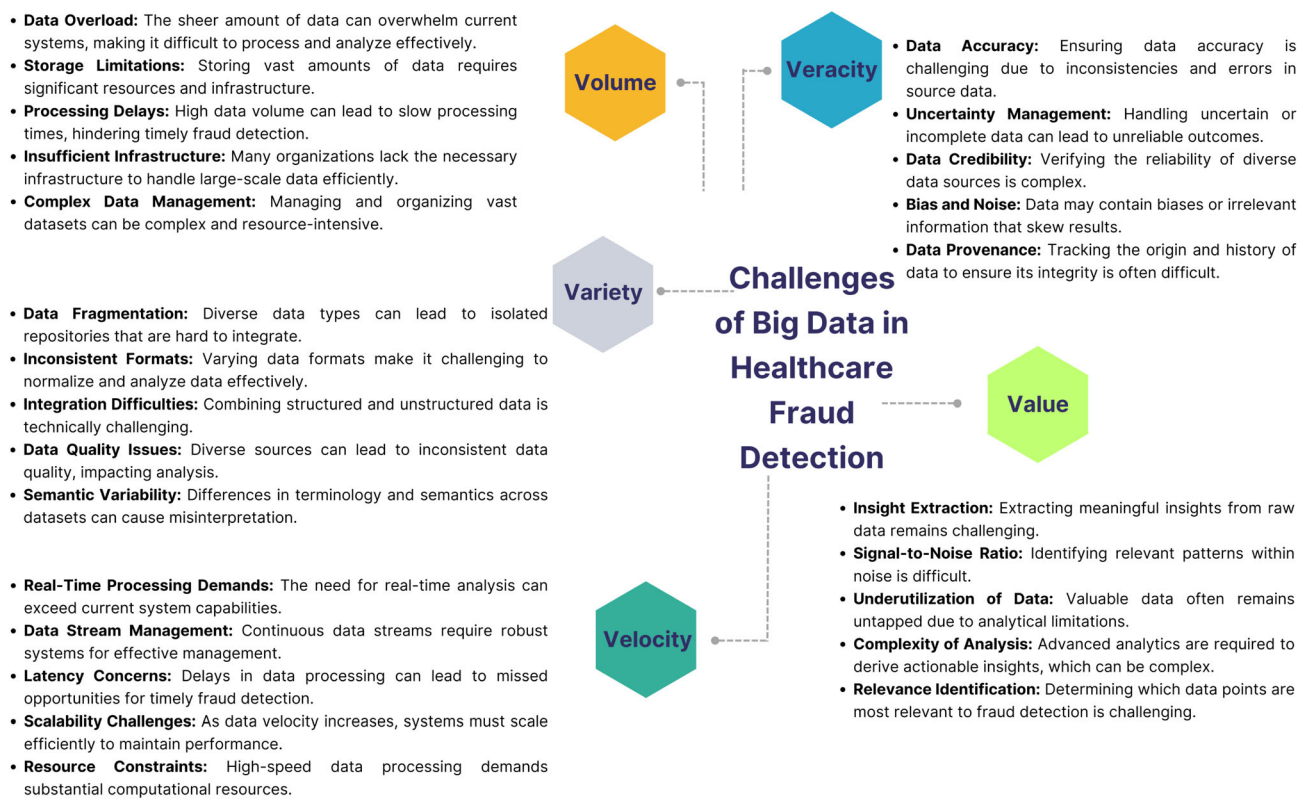

Fig. 1 Challenges of the 5Vs of Big Data in Healthcare Fraud Detection

- *Adopting advanced frameworks:* Utilizing Big Data technologies such as Apache Spark or Hadoop can significantly enhance processing power, allowing for real-time analysis and more precise fraud detection.

- *Developing scalable solutions:* Creating algorithms and systems that scale with data growth is crucial. This ensures detection mechanisms remain robust and efficient, even as healthcare data volumes increase.

- *Integrating diverse data sources:* Emphasizing the integration of diverse data types can provide a comprehensive understanding of fraud patterns, improving detection accuracy and reducing false positives.

7.2 Variety

The Variety aspect of Big Data in healthcare fraud detection involves diverse data types such as structured billing records, unstructured physician notes, and heterogeneous patient information. Effectively managing this diversity is crucial for creating comprehensive fraud detection models. Handling diverse data types and sources poses significant challenges. Integrating structured and unstructured data requires sophisticated data processing and normalization techniques. This diversity can lead to data silos, inconsistencies, and integration difficulties, which hinder comprehensive analysis and accurate fraud detection. The analysis of reviewed studies (Tables 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14) highlights several weaknesses:

1. *Inadequate handling of data variety:* Many studies (Tables 1, 2, 4, 6, 10, 11, 12, 13, and 14) fail to address the integration of diverse data types, limiting their ability to detect complex fraud patterns.
2. *Limited use of integrated frameworks:* Table 17 shows only a few studies (Tables 3, 5, 7, 8, and 9) effectively utilize frameworks to handle data variety. This restricts the depth of analysis and the richness of insights derived.
3. *Challenges in data normalization and integration:* The lack of effective data normalization and integration strategies leads to fragmented analyses, reducing the accuracy and reliability of fraud detection systems.

7.2.1 Implications for future research: variety

To overcome these challenges, future research should focus on:

- *Developing robust integration techniques:* Implementing advanced data integration and normalization techniques is vital for combining structured and unstructured data sources, enabling a holistic view of fraud patterns.
- *Leveraging advanced analytical tools:* Utilizing tools capable of processing diverse data types will enhance the ability to identify nuanced fraud patterns and improve detection accuracy.
- *Creating interdisciplinary approaches:* Collaboration between data scientists, healthcare professionals, and IT experts can facilitate the development of comprehensive models that effectively utilize data variety.

7.3 Velocity

The Velocity aspect of Big Data refers to the speed at which data is generated, processed, and analyzed. Efficient data processing is essential in healthcare fraud detection to promptly identify and prevent fraudulent actions. Efficiently managing the rapid flow of data necessitates the use of strong systems that can process information in real-time or almost real-time. The challenge lies in developing infrastructure and algorithms that can efficiently manage continuous data streams without compromising accuracy. An analysis of the reviewed studies (Tables 1 to 14) reveals several key weaknesses:

1. *Insufficient real-time processing:* All studies do not address the Velocity aspect, indicating a lack of focus on real-time data processing capabilities.
2. *Limited use of streaming technologies:* The review of literature shows a minimal adoption of technologies that support streaming data, which is essential for handling high-velocity data effectively.
3. *Challenges in scalability and responsiveness:* Existing systems often struggle with scalability and responsiveness, leading to delays in fraud detection and increased vulnerability to fraudulent activities.

7.3.1 Implications for Future Research: Velocity

To address these challenges, future research should focus on:

- *Implementing real-time processing frameworks:* Leveraging frameworks like Apache Kafka and Apache Storm can enhance capabilities for processing streaming data efficiently, enabling timely fraud detection.
- *Developing scalable algorithms:* Designing algorithms that maintain performance under high data velocity will ensure quick and accurate detection of fraudulent patterns.
- *Enhancing system responsiveness:* Building systems that can adapt to fluctuating data speeds will improve overall reliability and effectiveness in detecting fraud.

7.4 Veracity

The Veracity aspect of Big Data refers to the precision and reliability of data. Data integrity is of utmost importance in healthcare fraud detection to ensure accurate identification of fraudulent activities. Data veracity in healthcare refers to the process of managing discrepancies, mistakes, and uncertainties that naturally exist in data obtained from different sources. These problems can undermine the dependability of fraud detection systems, resulting in false positives and negatives. The analysis of the reviewed studies (Tables 1 to 14) highlights several points:

1. *Adequate focus on data accuracy:* Almost all studies performed comprehensive strategies to address data veracity, impacting the reliability of their findings.

2. *Challenges in managing data uncertainty:* Existing systems struggle to handle inconsistencies and uncertainties, which can lead to compromised fraud detection results.

7.4.1 Implications for future research: veracity

To overcome these challenges, future research should focus on:

- *Implementing robust data validation methods:* Utilizing advanced data cleaning and validation techniques can improve data quality, leading to more reliable fraud detection outcomes.

- *Developing techniques for handling uncertainty:* Creating methods to manage and mitigate data uncertainty will enhance the accuracy and trustworthiness of detection systems.

- *Fostering data source verification:* Establishing processes for verifying the credibility of data sources will help ensure the integrity of data used in fraud detection.

7.5 Value

The Value aspect of Big Data focuses on deriving actionable insights from data. In healthcare fraud detection, extracting meaningful value is essential for effective decision-making and fraud prevention. Transforming vast amounts of raw data into valuable insights involves complex analytical processes. The challenge lies in identifying relevant patterns and trends that can accurately indicate fraudulent activities while minimizing noise and irrelevant information. The analysis of the reviewed studies (Tables 1 to 14) highlights several points:

1. *Extensive insight extraction:* Almost all studies struggle to extract meaningful insights, resulting in underutilized data potential.

2. *Challenges in identifying relevant patterns:* Existing systems often face difficulties in pinpointing significant patterns, leading to inefficient fraud detection and increased false positives.

7.5.1 Implications for future research: value

To address these challenges, future research should focus on:

- *Implementing advanced analytics:* Utilizing machine learning and AI-driven analytics can enhance the ability to extract valuable insights from large datasets, improving fraud detection accuracy.

- *Focusing on data-driven decision making:* Developing systems that prioritize actionable insights will enable better decision-making and more effective fraud prevention strategies.

- *Enhancing pattern recognition capabilities:* Improving techniques to identify relevant patterns will reduce false positives and increase the efficiency of fraud detection systems.

8 Integration of AI and big data applications for healthcare fraud detection

The integration of AI and Big Data applications in healthcare fraud detection represents a significant advancement in addressing fraudulent activities within the industry. The eFAD suite, a framework developed for this purpose, is actor- and commodity-independent, allowing for adaptability in the dynamic environment of fraud detection [37]. This framework utilizes interactive machine learning (IML), which incorporates expert knowledge in an unsupervised setting, enhancing the detection of fraudulent and abusive cases in healthcare. Big Data analytics plays a crucial role in this integration by enabling the identification of fraud, abuse, waste, and errors in health insurance claims. These applications not only help in recovering hidden cost overruns but also improve patient care by reducing recurrent losses [38]. The ability to analyze large datasets for patterns and anomalies is a hallmark of AI in healthcare fraud detection, making it possible to uncover irregularities that traditional methods might miss. A specific methodology within this framework is claim anomaly detection, which focuses on identifying irregularities in health insurance claims. This approach is essential for leveraging AI and Big Data to enhance the accuracy and efficiency of fraud detection systems. Additionally, the use of Expectation Maximization (EM) for clustering similar actors within the fraud detection system further exemplifies the application of AI techniques in analyzing healthcare data. Moreover, the two-stage data warehousing approach is employed for proactive risk calculations, illustrating how Big Data methodologies can be integrated into healthcare fraud detection frameworks. This comprehensive approach not only addresses the fragmented nature of abnormal behaviors but also ensures that the system can adapt to new types of fraud and abuse as they emerge. The iWebCare project exemplifies another practical application of these technologies, having developed a flexible online fraud detection platform aimed at minimizing financial losses due to fraud in healthcare. This platform detects erroneous or suspicious records in healthcare datasets, promoting consistency and awareness across healthcare systems in the EU [39]. In conclusion, the

integration of AI and Big Data applications in healthcare fraud detection is transforming the landscape of fraud prevention, offering innovative solutions that enhance operational efficiencies and improve the overall quality of healthcare services. The combination of these technologies not only aids in identifying fraudulent activities but also fosters a more trustworthy healthcare environment for patients and providers alike.

8.1 Leveraging big data characteristics for enhanced healthcare fraud detection: an AI-powered approach

The convergence of Big Data and AI presents a transformative opportunity to revolutionize healthcare fraud detection. This section explores how AI can effectively leverage the inherent characteristics of Big Data, commonly referred to as the 5Vs, to enhance the accuracy, efficiency, and scalability of fraud detection systems, summarized in Figure 2:

- *Volume (massive data scale)*
  - *Challenge:* Healthcare generates enormous amounts of data (claims, patient records, provider information). Analyzing this vast volume efficiently to identify fraudulent patterns is computationally demanding.
  - *AI Solutions:*

    Distributed machine learning: Employing frameworks like Apache Spark to train AI models on massive datasets distributed across a cluster of machines, enabling faster processing and analysis.
    Deep Learning: Utilizing deep neural networks to automatically learn complex patterns and anomalies from massive volumes of data, improving fraud detection accuracy.
- *Velocity (data generation speed)*
  - *Challenge:* Healthcare data is generated rapidly. Real-time fraud detection is crucial to prevent significant financial losses.
  - *AI Solutions:*

    Stream processing: Implementing AI models within stream processing frameworks like Apache Kafka to analyze data in real-time as it's generated, enabling immediate fraud alerts.
    Online machine learning: Using algorithms that can continuously learn and adapt to new data and emerging fraud patterns in real-time, ensuring the system remains effective.
- *Variety (data diversity)*

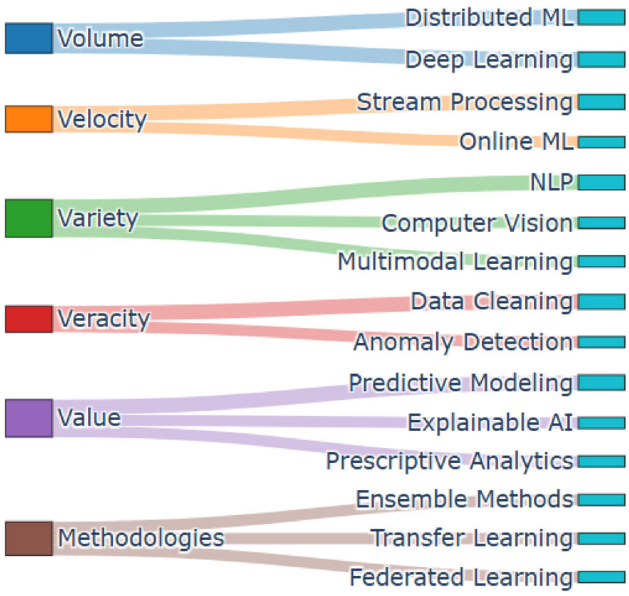

Fig. 2 Leveraging big data characteristics for enhanced healthcare fraud detection: an AI-powered approach

  - *Challenge:* Healthcare data comes in various formats (structured claims data, unstructured doctor's notes, medical images). Integrating and analyzing this diverse data is essential for a comprehensive view.
  - *AI Solutions:*

    Natural language processing (NLP): Extracting insights from unstructured text data like medical records and claim descriptions to identify inconsistencies or suspicious patterns.
    Computer vision: Analyzing medical images to detect anomalies or potential fraud in areas like unnecessary procedures or fabricated injuries.
    Multimodal learning: Combining different data types (e.g., claims data with medical images) to create a more holistic view and improve fraud detection accuracy.
- *Veracity (data trustworthiness)*
  - *Challenge:* Ensuring data quality and reliability is crucial for accurate fraud detection. Inconsistent or incomplete data can lead to false positives or negatives.
  - *AI Solutions:*

    Data cleaning and preprocessing: Using AI algorithms to identify and correct errors, inconsistencies, and missing values in the data, improving its reliability.

Anomaly Detection: Employing AI to identify outliers and unusual patterns that may indicate fraudulent activity or data quality issues.

- *Value (data usefulness)*
  - *Challenge:* Extracting meaningful insights from Big Data to inform fraud prevention strategies and improve healthcare processes.
  - *AI Solutions:*

    Predictive modeling: Developing AI models to predict future fraud risks based on historical data and identify vulnerable areas.
    Explainable AI (XAI): Using techniques to understand why an AI model flagged a particular case as fraudulent, providing transparency and insights for investigators.
    Prescriptive analytics: Using AI to recommend actions to prevent future fraud, such as targeted audits or policy changes.

- *Methodologies and approaches*
  - *Ensemble methods:* Combining multiple AI models (e.g., decision trees, support vector machines) to improve prediction accuracy and robustness.
  - *Transfer learning:* Leveraging pre-trained AI models on large datasets and adapting them for specific healthcare fraud detection tasks, reducing training time and data requirements.
  - *Federated learning:* Training AI models across multiple institutions without sharing sensitive patient data, enabling collaborative fraud detection while preserving privacy.

By effectively addressing the 5Vs of Big Data with these AI-driven solutions, healthcare fraud detection systems can achieve significant improvements in accuracy, efficiency, and scalability, ultimately contributing to more robust fraud prevention and resource protection within the healthcare industry.

9 Research directions for 5V's challenges in healthcare fraud detection

The exploration of Big Data in healthcare fraud detection has yielded promising results, but there remain significant avenues for future research to further enhance the accuracy, efficiency, and scalability of these systems. Here are some key research directions to address the challenges posed by the 5Vs of Big Data, summarized in Figure 3:

- *Volume*
  1. *Efficient processing of massive claims data:* Develop and implement scalable algorithms for handling massive volumes of healthcare claims data, enabling efficient identification of fraudulent patterns and anomalies.
  2. *Optimized storage for diverse healthcare data:* Investigate cloud-based and distributed storage solutions to accommodate the increasing volume and variety of healthcare data, ensuring cost-effectiveness and accessibility.
- *Variety*
  1. *Advanced integration of heterogeneous data:* Develop robust data integration techniques to combine diverse healthcare data sources, such as claims data, patient records, and provider information, for a comprehensive view of potential fraud.
  2. *AI-driven analysis of unstructured data:* Explore the use of AI and NLP to extract meaningful insights from unstructured healthcare data, such as clinical notes and medical images, to enhance fraud detection capabilities.
- *Velocity*
  1. *Real-time fraud detection:* Implement real-time analytics platforms, like Apache Kafka or Flink, to enable immediate detection and prevention of fraudulent activities as they occur, minimizing financial losses and protecting patients. *Edge Computing for Timely Fraud Prevention:* Investigate the

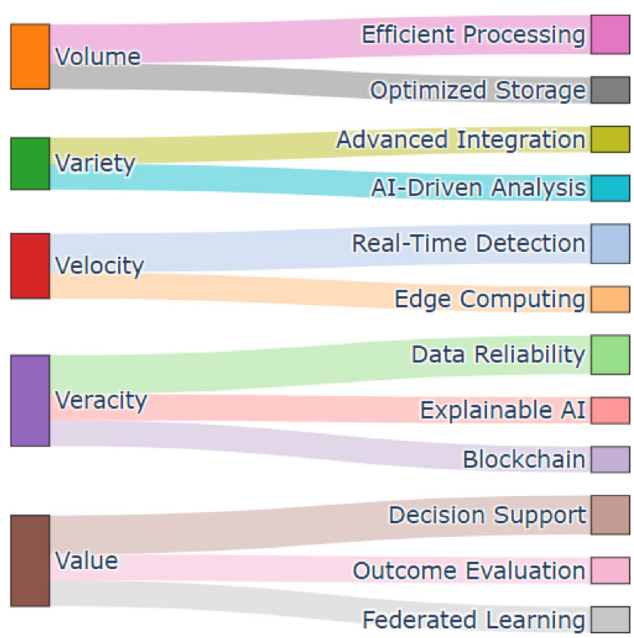

Fig. 3 Challenges of the 5Vs of Big Data in Healthcare Fraud Detection - Research Directions

application of edge computing to process data closer to the source, reducing latency and enabling more timely identification and prevention of fraudulent activities.

- *Veracity*
  1. *Enhancing data reliability and trustworthiness:* Develop advanced data validation techniques and uncertainty management models to ensure the accuracy and reliability of healthcare data used for fraud detection, minimizing false positives and negatives.
  2. *Explainable AI for transparent fraud detection:* Utilize explainable AI (XAI) to provide insights into the decision-making process of AI-based fraud detection models, increasing transparency, trust, and accountability.
  3. *Secure data management with blockchain:* Explore the use of blockchain technology to create a secure and transparent system for managing healthcare data, ensuring data integrity and reducing the risk of fraud and data breaches.
- *Value*
  1. *Data-driven decision support for fraud prevention:* Develop systems that provide actionable insights and decision support to healthcare professionals, enabling proactive fraud prevention strategies and efficient resource allocation.
  2. *Outcome-based evaluation of fraud detection systems:* Evaluate the effectiveness of fraud detection systems based on their impact on healthcare outcomes, such as reducing financial losses, improving patient safety, and enhancing the overall quality of care.
  3. *Collaborative fraud detection with federated learning:* Investigate the use of federated learning to enable collaborative fraud detection across multiple healthcare institutions without compromising data privacy, enhancing the accuracy and generalizability of models.

By focusing research efforts on these specific areas, the potential of Big Data in enhancing healthcare fraud detection can be fully realized, leading to more efficient, reliable, and effective systems that protect patients and preserve healthcare resources.

10 Concrete suggestions for 5V's challenges in healthcare fraud detection

In the realm of healthcare fraud detection, effectively managing the 5V's of Big Data is critical to enhancing the accuracy and efficiency of detection systems. Each V presents unique challenges that require targeted strategies and cutting-edge technologies to address. By implementing robust solutions for each aspect, healthcare organizations can improve data handling, ensure data quality, and extract meaningful insights, ultimately leading to more effective fraud prevention and resource optimization. Figure 4 addresses the concrete suggestions for overcoming the Big Data 5Vs challenges in credit risk assessment.

10.1 Volume

In the context of healthcare fraud detection, managing the vast *Volume* of Big Data is essential for effective analysis and processing. The ability to handle large datasets is crucial for identifying patterns and anomalies that indicate fraudulent activities. To achieve this, several strategies can be employed:

1. Adopt scalable cloud solutions: utilize cloud-based platforms such as Amazon Web Services (AWS), Google Cloud Platform (GCP), or Microsoft Azure to leverage scalable storage and computing resources. These platforms offer flexibility and scalability to handle large datasets without significant upfront infrastructure investment.
2. Implement distributed computing frameworks: use frameworks like Apache Hadoop and Apache Spark for parallel processing and distributed storage. These tools enable efficient data handling by dividing large datasets across multiple nodes, reducing processing time and enhancing performance.
3. Optimize data storage with advanced techniques: employ data compression algorithms and columnar storage formats like Apache Parquet or Apache ORC. These techniques reduce storage requirements and improve read/write efficiency, crucial for handling large volumes of data.
4. Leverage data lakes for centralized storage: establish data lakes to store raw, unprocessed data in its native format. This approach allows for flexible data retrieval and processing, accommodating various data types and structures.
5. Utilize efficient data ingestion tools: use tools like Apache NiFi or Kafka for streamlined data ingestion pipelines. These tools facilitate the efficient movement of large volumes of data from diverse sources into

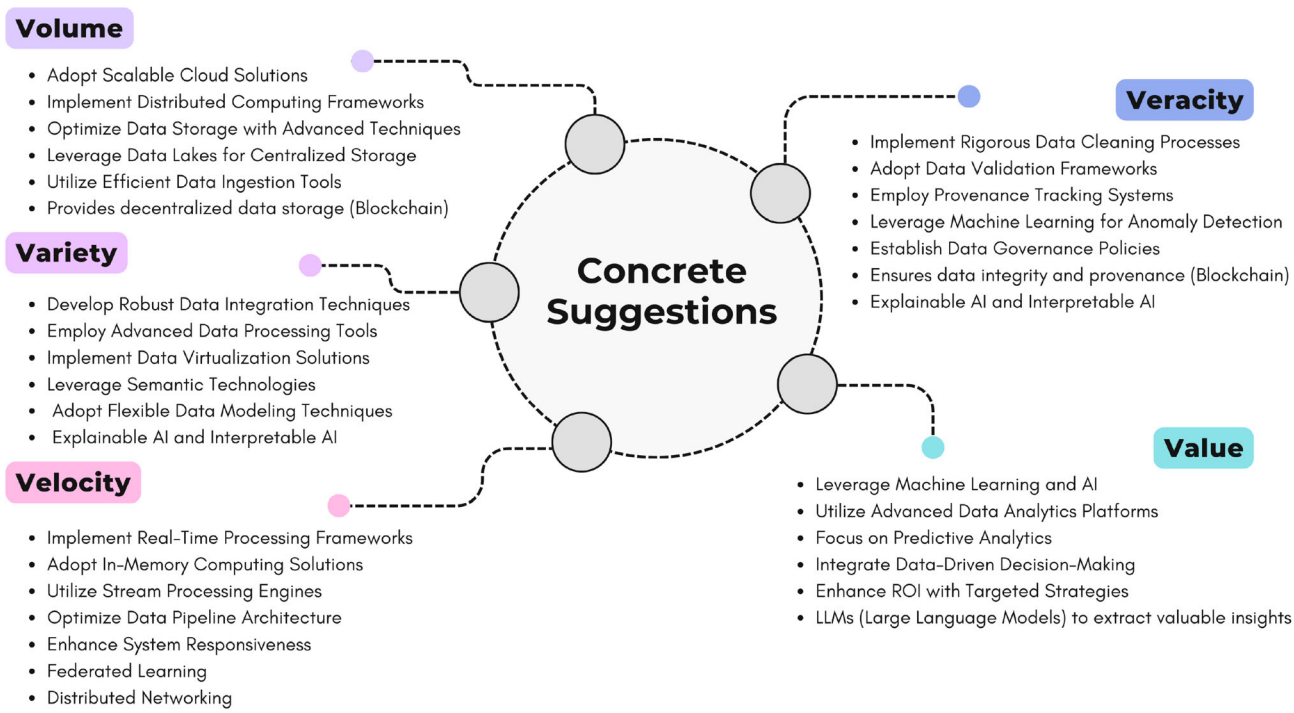

Fig. 4 Big data 5 V's discrete suggestions for healthcare fraud detection

processing systems, ensuring data is readily available for analysis.

6. Blockchain: provides decentralized data storage, enhancing security and traceability, but may face scalability challenges with large datasets.
7. Distributed networking: facilitates scalable data processing by distributing workloads across multiple nodes, making it highly feasible for handling large volumes.

10.2 Variety

In healthcare fraud detection, managing the *Variety* of Big Data is critical for effectively integrating and analyzing diverse data types. This involves handling structured, semi-structured, and unstructured data from numerous sources. To address these challenges, the following strategies are essential:

1. Develop robust data integration techniques: Use ETL (Extract, Transform, Load) tools like Apache Nifi or Talend to seamlessly integrate structured and unstructured data from various sources, ensuring consistency and compatibility for analysis.
2. Employ advanced data processing tools: utilize natural language processing (NLP) tools such as spaCy or NLTK to analyze unstructured text data from clinical notes or patient records. These tools can extract valuable insights and patterns from textual information.
3. Implement data virtualization solutions: use data virtualization platforms like Denodo or Red Hat JBoss to access and integrate data without physical consolidation. This allows real-time integration of data from different sources and formats.
4. Leverage semantic technologies: apply semantic web technologies and ontologies to provide a unified framework for data from diverse domains. This enhances data interoperability and facilitates more meaningful analysis.
5. Adopt flexible data modeling techniques: implement NoSQL databases such as MongoDB or Cassandra to store and manage semi-structured or unstructured data. These databases offer flexibility in handling diverse data types and structures.
6. Explainable AI and interpretable AI: enhance understanding of diverse data patterns and model decisions, promoting trust and transparency.

10.3 Velocity

In healthcare fraud detection, effectively managing the *Velocity* of Big Data is crucial for processing information in real-time or near real-time. This capability is essential for promptly identifying and responding to fraudulent activities. Key strategies include:

1. Implement real-time processing frameworks: use Apache Kafka and Apache Flink for real-time data

streaming and processing. These tools enable the ingestion and analysis of high-speed data flows, crucial for timely fraud detection.
2. Adopt in-memory computing solutions: leverage in-memory data grids like Apache Ignite or Hazelcast to reduce latency and improve the speed of data processing. This allows for faster access to data, enhancing real-time analysis capabilities.
3. Utilize stream processing engines: employ stream processing engines such as Apache Storm or Spark Streaming to handle continuous data streams. These engines can process large volumes of data on-the-fly, providing immediate insights.
4. Optimize data pipeline architecture: design efficient data pipelines using tools like Apache Beam to streamline data flow and ensure minimal delays between data ingestion and processing.
5. Enhance system responsiveness: implement load balancing and auto-scaling techniques to dynamically adjust resources based on data velocity, ensuring consistent performance and responsiveness.
6. Federated learning: enables real-time processing by distributing learning across devices without centralizing data, maintaining speed and privacy.
7. Distributed networking: supports rapid data transfer and processing, ensuring timely analysis and response to high-velocity data streams.

10.4 Veracity

In healthcare fraud detection, managing the *Veracity* of Big Data is essential for ensuring data accuracy and reliability. This involves implementing strategies that enhance data quality and trustworthiness:

1. Implement rigorous data cleaning processes: use tools like OpenRefine or Trifacta to clean and preprocess data, removing inaccuracies and inconsistencies to enhance data quality.
2. Adopt data validation frameworks: utilize validation libraries such as Apache Beam or Deequ to systematically verify data integrity and correctness across datasets.
3. Employ provenance tracking systems: implement data lineage tools like Apache Atlas to track data origins and transformations, ensuring transparency and accountability.
4. Leverage machine learning for anomaly detection: use machine learning algorithms to identify and rectify anomalies or outliers in data, improving overall data reliability.
5. Establish data governance policies: develop comprehensive data governance frameworks to enforce data quality standards, ensuring consistent data management practices.
6. Blockchain: ensures data integrity and provenance, making it feasible for maintaining data veracity.
7. Explainable AI and interpretable AI: improve data reliability by providing insights into model behavior and detecting anomalies.

10.5 Value

In healthcare fraud detection, maximizing the *Value* of Big Data involves extracting actionable insights to enhance decision-making and resource allocation. Key strategies include:

1. Leverage machine learning and AI: implement advanced machine learning models and AI algorithms to identify patterns and trends, enhancing the ability to detect complex fraud scenarios.
2. Utilize advanced data analytics platforms: use platforms like Tableau (such as via Live Connections to big data sources, Hyper Engine, or Integration with Big Data platforms) or Power BI (such as via DirectQuery, Azure Integration, or Data Reduction techniques) to visualize data insights, making them accessible and actionable for decision-makers.
3. Focus on predictive analytics: employ predictive analytics tools to forecast potential fraud activities, enabling proactive fraud prevention measures.
4. Integrate data-driven decision-making: develop systems that prioritize actionable insights and key performance indicators (KPIs), ensuring efforts are aligned with strategic objectives.
5. Enhance ROI with targeted strategies: implement targeted fraud detection strategies that focus on high-value insights, improving the return on investment and resource allocation.
6. LLMs (large language models): extract valuable insights from unstructured data, enhancing the analytical depth and precision.

11 Conclusion

This review highlights the transformative potential of Big Data analytics in modernizing healthcare fraud detection. We offer a clear roadmap to effectively harness Big Data, driving meaningful change and addressing emerging challenges in healthcare fraud detection. This review analyzes how the 5Vs can be leveraged to improve fraud detection systems, identifies existing research gaps, and proposes future research directions. Furthermore, we address the

integration of AI and Big Data applications in healthcare fraud detection, highlighting their role in enhancing operational efficiencies and improving the overall quality of healthcare services.

Author contributions Amin conducted the primary research. Both authors contributed equally to the data collection, findings, and discussions throughout this research.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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